

Ben Snell

Creative Software Engineer

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EDUCATION

Carnegie Mellon University B.A. Experiential Art & Design

Aug 2015 - Dec 2017

4.0 GPA + Honors

Self-defined major to develop interactive tools and systems across virtual and tangible media.

Select classes:

- Intro to Machine Learning
- Matrices and Linear Transforms
- Human-Robot Interaction
- Sensing and Sensors
- Computational Perception
- Computing for Creative Practices
- Speculative Critical Design

SKILLS

Generative 2D/3D Graphics
Computer Vision
Data Processing
Generative AI
Algorithm Design
Computational Design
Innovation Engineering
Visual Application Development

TOOLS

C++, OpenCV, OpenGL, GLSL
Python, PyTorch
Javascript, React, WebGL, Three.js
Node.js, Electron
Bash, Docker, AWS, Linux
Unity, OpenFrameworks
Git, Figma, Asana

SUMMARY

Software Engineer with 9 years of experience in generative graphics, computer vision and computational design. Self-starter, versatile, resourceful. Thrives in collaborative, cross-disciplinary environments, tackling ambiguous problem sets with positive social impact.

EXPERIENCE

Educational Sabbatical

Feb 2023 - Jan 2024

- Developed generative NFT orchids using 3D morphable models in Javascript, p5.js, WebGL and GLSL shaders.
- Made generative AI sculptures using a VAE trained on 3D meshes with PyTorch.

Senior R&D Software Engineer

Amazon, Alexa Devices

July 2021 - Jan 2023

- Developed data processing pipelines from 3D scanning to reconstruction to templization.
- Built statistical shape models for human factors research using MATLAB and Python.
- Architected & coded a 3D graphics web app to help designers evaluate wearables' fitness using React, Three.js, Node.js, Docker and AWS.

Creative Software Engineer

Local Projects

Jan 2019 - July 2021

- Led concept through production of interactive museum exhibits using OpenFrameworks, Unity (C#), Docker, Node.js, React, Python, IOT/RPi and sensors (LiDAR, Kinect, etc.)
- Key collaborator across UX / experiential / industrial design, CI / CD and AV integration.
- Developed object detection & tracking and LSTM signal extraction algorithms for active mocap systems in C++.
- Applied AI to novel data visualization techniques using GANs, UMAP and TensorFlow.

Innovator in Residence

Autodesk

Feb - June 2018

- Applied advanced manufacturing methods to CNC robotics & digital fabrication using Fusion 360.

Software Developer Intern

Troika

June - Aug 2017

- Developed robotic mocap software using C++ and C4D.

Software Developer

ATONATON Robotics

July - Aug 2016

- Programmed the vision and decision-making system of an interactive industrial robotic arm using C++, OpenCV and OpenFrameworks.

Interaction Design Intern

IDEO

May - Aug 2016

- Conducted design research and made physical computing prototypes using Arduino / Raspberry Pi.

Junior Creative Technologist

72andSunny

Sep 2014 - July 2015

- Designed UX and prototyped AR / XR for brand campaigns.